OCR routing report, 21 October 2026

Source: `merged/final/MERGED_REPORT.md`, produced by `tools/merge_results.py --round2` from the desktop export (22,731 rows) and the validated laptop export (18,244 rows, SHA-256 `ebc69a53...f5da753`, matches). Merged: 40,975 rows, 26 same-model comparison groups. Both exports passed `check_conformance.py --round2` (CONFORMANT). Everything below is scores and counts from code. No model output was read by a person or by Claude. Failed cases count as zero.

Routing choices

Task	Route	Measured basis
Crop OCR (prose crops)	Qwen3-VL 8B instruct, baseline	Character error 1.2% vs 2.0% for Qwen2.5-VL 7B (50 crops each, laptop). 8B matches the desktop 32B models (1.1% to 1.2%) and runs in 2.6 s median on the laptop
Charts, question answering	Qwen3-VL 8B instruct, **self-check off**, baseline or upscale	79.5% baseline (200 cases, laptop). Self-check drops it to 36.0%, a loss of 43 points (95% CI -51 to -35)
Charts, 7B fallback	Qwen2.5-VL 7B, **self-check on**	52.5% to 79.0%, a gain of 26.5 points (CI +18.5 to +34), and +17.8 points on the 1,322-case v2 confirmation set
Chart to table	8B with self-check or expert prompt	Score 0.42 to 0.93 (self-check), 60 cases. Baseline 8B is poor here
Tables	Any of the 8B variants, baseline is enough	TEDS 0.82 to 0.83, no variant differs significantly
Whole-page text masking (layoutmask)	**Experimental, not routed by default**	Helps 7B (+0.085) and the desktop 32B (+0.067); no gain on 8B (+0.010, CI includes 0). It lowers numeric recall on 32B (0.909 to 0.720)
Three-vote chart answers (vote3)	Use only with 7B or 32B	+0.045 on 7B (CI +0.030 to +0.061), +0.026 on 32B; no effect on 8B (-0.002, not significant)

Charts, question answering (relaxed accuracy, 200 development cases)

Model	Baseline	Expert	Self-check	Upscale
Qwen2.5-VL 7B, laptop	52.5%	47.5%	**79.0%**	47.5%
Qwen2.5-VL 7B, desktop	53.5%	44.5%	80.0%	49.0%
Qwen3-VL 8B instruct, laptop	**79.5%**	75.0%	36.0%	80.0%
Qwen3-VL 8B instruct, desktop	78.0%	76.0%	32.5%	79.5%
Qwen3-VL 32B, desktop (reference)	85.0%	84.0%	91.0%	89.5%

Round-2 confirmation (1,322 cases, laptop), change from baseline with 95% CI:

Model	Variant	Change	95% CI	Verdict
7B	vote3	+0.045	+0.030 to +0.061	better (confirmatory)
7B	self-check	+0.178	+0.150 to +0.207	better (supporting)
7B	upscale	-0.038	-0.055 to -0.021	worse
8B	vote3	-0.002	-0.011 to +0.008	not significant
8B	self-check	-0.427	-0.455 to -0.396	worse
8B	upscale	+0.015	+0.003 to +0.027	slightly better
32B, desktop	vote3	+0.026	+0.014 to +0.037	better (confirmatory)

Crop OCR (character error, lower is better, 50 crops)

Model	Laptop	Desktop
Qwen3-VL 8B instruct	**0.012**	0.012
Qwen2.5-VL 7B	0.020	0.018
32B-class models (desktop only)	n/a	0.011 to 0.012

Tables (TEDS, 60 cases) and chart data extraction

Model	Table TEDS laptop	Table TEDS desktop	Chart to table laptop (baseline / self-check)
Qwen3-VL 8B instruct	0.818	0.819	0.418 / 0.930
Qwen2.5-VL 7B	0.724	0.724	0.650 / 0.626

Page masking (word F1)

Model	Cases	Baseline	Layoutmask	Change [95% CI]
7B, laptop	80	0.757	0.843	+0.085 [+0.039, +0.135]
8B, laptop	80	0.828	0.838	+0.010 [-0.035, +0.050]
32B, desktop	80	0.814	0.881	+0.067 [+0.033, +0.101]

Do the two machines agree?

Qwen2.5-VL 7B has the same weights on both machines. Of its 13 comparisons, 11 reproduced; 2 disagreed (chart_qa expert, +3.0 points; chart_table upscale, -14.8 points, a 60-case task). Crop OCR and tables reproduced.

Caveats

- The 8B digest on the laptop (`70e58e37...b96eb`) differs from the desktop record. The merged report marks all 8B rows "different weights: not a reproduction test". The 8B comparison is a method comparison, not a reproduction. Even so, laptop and desktop 8B scores agree closely (for example crop error 0.0123 vs 0.0124).
- 7B and 8B results are not reproductions of the desktop 32B. The vote3 pick was made on 32B development data. On 7B and 8B it replicates the protocol with a different model, and does not count as a new selection.
- Many tasks have only 50 to 60 cases, so their intervals are wide. Holm-adjusted p-values are in the merged report. Only the round-2 confirmation set (1,322 charts, 80 pages) is large.
- Page masking stays experimental per the round-2 contract, and needs a numeric-recall check before use.
- Self-check helps weaker models and harms the stronger ones on this prompt. Route it per model, not globally.